

Aasim Akhtar

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Professional Summary

Backend Engineer and MCA graduate with hands-on experience building scalable and secure RESTful APIs using Node.js, Express.js, MongoDB, PostgreSQL, and Redis. Experienced in designing production-ready systems with JWT authentication, role-based access control, caching, rate limiting, transactions, and payment integrations. Built and deployed backend projects including a double-entry Ledger API, Real-Time Leaderboard, E-Commerce Backend, Smart Link API, and Voting Management System. Strong focus on backend architecture, performance optimization, database design, and building reliable systems ready for real-world production environments..

Education

Master of Computer Applications (MCA) — GNIOT, Greater Noida 2024 – 2026
CGPA: 8.72

Bachelor of Computer Applications (BCA) — Raj Narayan College, Hajipur 2021 – 2023
CGPA: 8.56

Technical Skills

Language: C++

Backend: Node.js, Express.js, REST API design, JWT, bcrypt, Nodemailer, Multer, Cloudinary

Frontend: React.js, JavaScript, HTML5, CSS3

Databases: MongoDB, PostgreSQL, Redis

Tools: Docker, Render, Vercel, Cloudinary, dotenv, Git, GitHub, Postman, VS Code

Projects

Ledger API | Node.js · Express · MongoDB · Render · Nodemailer · [Live Demo](#) 2026 – Present

- Built a secure ledger backend using Node.js, Express, MongoDB, and Mongoose, supporting user authentication, account management, and financial transactions.
- Designed an immutable double-entry ledger system that calculates account balances from credit and debit records rather than storing mutable balances.
- Implemented JWT-based authentication with HTTP-only cookies, token blacklisting on logout, bcrypt password hashing, and role-based system-user authorization.
- Developed idempotent, MongoDB transaction-based fund transfers with account-status and balance validation, plus automated email notifications through Gmail OAuth.

Real-Time Leaderboard API | Node.js · Express · MongoDB · Redis · Render · [Live Demo](#) 2025 – 2026

- Built a Real-Time Leaderboard API using Node.js, Express.js, MongoDB, and Redis Sorted Sets for fast global and per-game rankings.
- Implemented JWT authentication, Redis token blacklisting, and login rate limiting for secure authentication and logout.
- Designed a dual-write architecture to store score history in MongoDB while maintaining live leaderboards in Redis.

- Reduced leaderboard response time from 1.53 s to 107 ms by replacing MongoDB aggregation with Redis Sorted Sets, sustaining 200 req/s with 0% request failures during Artillery load testing.

E-Commerce Backend | *Node.js · Express · MongoDB · Cloudinary · Razorpay · Stripe*

2024 - 2025

- Built a RESTful E-Commerce backend with Node.js, Express 5, and MongoDB, covering product catalog management, cart operations, and order processing across admin and customer roles.
- Integrated Cloudinary and Multer for multi-image product uploads, handling up to 4 images per product with asynchronous parallel uploads via Promise.all.
- Implemented Stripe Checkout for online payments alongside a Cash-on-Delivery flow, with a webhook-free verification endpoint that confirms payment status and clears the user's cart on success.
- Designed role-based JWT authentication separating customer-facing routes from admin-only product and order management endpoints (add/remove products, update order status, view all orders).

Smart Link API | *Node.js · Express · MongoDB · Redis*

2024

- Built a URL Shortener API with Node.js, Express 5, and MongoDB, generating collision-free Base62 short codes by encoding each document's ObjectId, eliminating the need for a separate uniqueness-check-and-retry loop.
- Implemented Redis-backed session authentication using express-session and connect-redis, replacing stateless JWTs with server-side revocable sessions for login/logout.
- Added per-IP rate limiting (10 requests/minute) on link-creation endpoints using Redis INCR/EXPIRE counters to prevent abuse and shortening spam.
- Designed a click-tracking system that atomically increments visit counts on redirect (findOneAndUpdate with \$inc) and exposes a per-link analytics endpoint scoped to the authenticated owner.

Voting Management System | *Node.js · Express · MongoDB · Redis*

2023

- Built a RESTful Election Management API using Node.js, Express 5, and MongoDB, implementing role-based access control (admin/voter) with JWT authentication to secure election, candidate, and voting endpoints.
- Engineered a secure OTP-based email verification system using Nodemailer and bcrypt-hashed OTPs, with Redis-backed rate limiting to cap OTP requests to 3 per 60 seconds and prevent abuse.
- Designed MongoDB schemas with compound unique indexes to enforce one-vote-per-user-per-election at the database level, eliminating race conditions in concurrent vote submissions.
- Reduced repeated database load on election/candidate listing endpoints by implementing Redis caching (5-minute TTL) with automatic cache invalidation on create/update/delete operations.